

Tutorial Sheet 9 solution

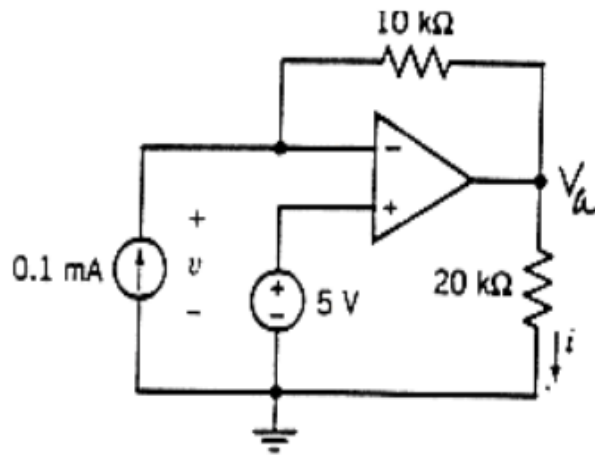
Solution 1(a)

$$\frac{v_o}{v_i} = -\frac{R_f}{R_1} = -\frac{25}{10} = -2.5$$

Solution 1(b)

$$\frac{v_o}{v_i} = \left(1 + \frac{5k}{10k}\right) = \left(1 + \frac{1}{2}\right) = \frac{3}{2}$$

Solution 2



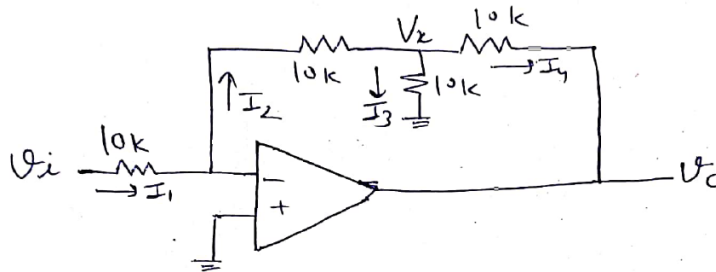
$$v = 5 \text{ V}$$

$$-\left(\frac{v_a - 5}{10000}\right) - 0.1 \times 10^{-3} - 0 = 0$$

$$\Rightarrow v_a = 4 \text{ V}$$

$$i = \frac{v_a}{20000} = \frac{1}{5} \text{ mA}$$

Solution 3



$$I_1 = I_2$$

$$\frac{v_i - 0}{10k} = \frac{0 - v_o}{10k} \Rightarrow v_{in} = -v_o \quad I$$

$$I_2 = I_3 + I_4$$

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$$\frac{0 - v_x}{10k} = \frac{v_x - 0}{10k} + \frac{v_x - v_o}{10k}$$

$$v_x = 2v_x - v_o$$

$$v_o = v_x = -v_{in} \quad ($$

$$\boxed{\frac{v_o}{v_{in}} = -1}$$

Solution 4

Applying KCL at node a ,

$$\frac{v_a - v_o}{40} = \frac{6 - v_a}{20}$$

$$v_a - v_o = 12 - 2v_a \quad \Rightarrow \quad v_o = 3v_a - 12$$

But $v_a = v_b = 2 \text{ V}$ for an ideal op amp, because of the zero voltage drop across the input terminals of the op amp. Hence,

$$v_o = 6 - 12 = -6 \text{ V}$$

Solution 5

$$-\left(\frac{v_o - 0}{3000}\right) - \left(\frac{12 - 0}{4000}\right) - 2 \cdot 10^{-3} = 0$$

$$\Rightarrow v_o = -15 \text{ V}$$

$$i_o + \frac{v_o}{6000} + \frac{v_o}{3000} = 0 \Rightarrow i_o = 7.5 \text{ mA}$$

